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Beyond neighborhood disadvantage: Local resources, green space, pollution, and crime as residential community correlates of cardiovascular risk and brain morphology in midlife adults

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Abstract

Objective: Residing in communities characterized by socioeconomic disadvantage confers risk for cardiometabolic diseases. Residing in disadvantaged communities may also confer risk for neurodegenerative brain changes via cardiometabolic pathways. This study tested whether features of communities - apart from conventional socioeconomic characteristics - relate not only to cardiometabolic risk, but also relative tissue reductions in the cerebral cortex and hippocampus.

Methods: Participants were 699 adults aged 30–54 years (340 women; 22.5% non-white) whose addresses were geocoded to compute community indicators of socioeconomic disadvantage, as well as air and toxic chemical pollutant exposures, homicide rates, concentration of employment opportunities, land use (green space), and availability of supermarkets and local resources. Participants also underwent assessments of cortical and hippocampal volumes and cardiometabolic risk factors (adiposity, blood pressure, fasting glucose, and lipids).

Results: Multilevel structural equation modeling demonstrated that cardiometabolic risk was associated with community disadvantage ($\beta = 0.10$, 95%CI = 0.01 to 0.18), as well as chemical pollution ($\beta = 0.11$, 95%CI = 0.02 to 0.19), homicide rates ($\beta = 0.10$, 95%CI = 0.01 to 0.18), employment opportunities ($\beta = -0.16$, 95%CI = −0.27 to −0.04), and green space ($\beta = -0.12$, 95%CI = −0.20 to −0.04). Moreover, cardiometabolic risk indirectly mediated the associations of

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Conflicts of Interest

The authors have no known conflicts to report.

Availability of data

Data may be made available on request by the study authors via completion of a Data Use Agreement (https://www.osp.pitt.edu/osp-teams/clinical-corporate-contract-services/negotiations/data-use-agreements-duas).

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several of these community features and brain tissue volumes. Some associations were nonlinear, and none were explained by participants' individual-level socioeconomic characteristics.

Conclusions: Features of communities other than conventional indicators of socioeconomic disadvantage may represent nonredundant correlates of cardiometabolic risk and brain tissue morphology in midlife.

Keywords

cerebral cortex; health inequalities; hippocampus; neighborhood disadvantage; socioeconomic position

INTRODUCTION

Residing in underserved communities that are characterized by social and economic disadvantage confers risk for chronic illnesses that lead to premature death (1–4). The chronic illnesses that are most unequally distributed across residential communities are cardiovascular and metabolic (cardiometabolic) diseases, such as ischemic heart disease and diabetes (5–14). The residential patterning of chronic illnesses and premature death has been attributed to features of communities that appear to confer more proximal risk for preclinical and pathogenic changes in systemic physiology (e.g., as reflected in components of the metabolic syndrome) and behaviors that are detrimental to health (2, 5, 9, 10, 12, 14). The latter features include socioeconomic (e.g. concentrated poverty), physical (e.g., land usage, proximity to resources) and social (e.g., crime and safety) aspects of residential environments that reflect upstream structural influences and inequitable policies that result in the nonrandom distribution of resources and exposures across communities (2, 6, 12).

Social, physical, and economic disadvantages of communities may not only confer risk for chronic illnesses that affect peripheral organ systems, such as the cardiovascular system, but also for morphological changes in the brain in midlife that precede neurocognitive decline and some dementias in later life (15–27). Indicators of community-level socioeconomic disadvantage, for example, have been related to neuroimaging measures of reduced brain tissue volume in midlife adults, particularly in the cerebral cortex and, somewhat less consistently, in the medial temporal lobe (17, 20). Moreover, in the latter studies, brain tissue changes observed in association with community-level disadvantage were partly explained (statistically mediated) by corresponding variation in biological and behavioral risk factors for cardiometabolic conditions. These neuroimaging findings agree with those from postmortem research demonstrating that community-level disadvantage associates with a greater severity of neurofibrillary tangles, neuritic plaque staging, and cerebral amyloid angiopathy seen on autopsy (19). Collectively, these findings agree with other lines of evidence indicating that elevated cardiometabolic risk predicts future cognitive decline and some dementias, such as vascular dementia and Alzheimer's disease, which themselves associate with socioeconomic disadvantage (28–31).

Notwithstanding existing evidence, indicators of community-level disadvantage used in the literature thus far have been largely constrained to aggregate measures of area-level income and education levels, housing patterns, and employment rates. The latter composite

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measures have in turn been associated with presumptive brain correlates of neurocognitive risk and associated variation in cardiometabolic risk factors (e.g., 17, 20, 21). A knowledge gap, however, is whether brain tissue changes and cardiometabolic risk factors relate to more specific features of communities that, according to social epidemiological models, more directly reflect the *physical* and *social* environments of residential areas, as well as access to local resources (e.g., 2). If so, this may provide a more comprehensive account of the community features, beyond relative socioeconomic disadvantage, that relate to cardiometabolic risk, neurocognitive aging, and brain health.

More precisely, *physical* features of residential communities include exposures to air and toxic chemical pollutants, access to healthy foods, and qualities of the built environment reflecting land use and green space (2, 12, 14). *Social* features include levels of crime and threats to safety, as well as access to employment opportunities and healthcare. In addition to more proximal or direct effects on risk (e.g., exposure of lung, vascular, and other organ tissues to ambient pollutants), these community-level features may indirectly affect risk (e.g., by constraining regular access to foods for a nutritious diet, health services, and safe engagement in physical activity) (2, 12, 14). What is more, such features may vary across communities independently of crude indicators of community-level disadvantage, such as the Area Deprivation Index (32) or other Census-based indicators of income, education, housing patterns, and overall employment rates (2). Put differently, diverse community features are not interchangeable with nonspecific indicators of 'disadvantage'. In these regards, research has yet to establish whether physical and social features of residential communities relate — independently of aggregate indicators of community-level disadvantage — to cardiometabolic risk factors for disease that in turn relate to morphological changes in the brain (i.e., reduced tissue volume) that may plausibly presage risk for later neurocognitive decline or dementias.

To address the latter possibility, this study examined how residential community features relate to presumptive brain correlates of neurocognitive aging via cardiometabolic risk pathways by addressing three aims. First, it examined whether physical and social environmental features (air and chemical pollutant exposures, homicide rates, availability of healthy foods, green space, employment opportunities, and local resources) are associated with cardiometabolic risk, above-and-beyond a conventional Census-based marker of community socioeconomic disadvantage. Second, it tested whether cardiometabolic risk associates with reductions in cortical and hippocampal tissue volumes. Lastly, to extend prior findings (17, 20) it tested cardiometabolic risk as a candidate biological and indirect pathway by which community physical, social, and socioeconomic features relate to relative reductions in brain tissue volumes across individuals.

Methods

Cross-sectional data were drawn from two cohort studies: Phase 2 of the Adult Health and Behavior project (AHAB-2; N = 490; 53% women; age range = 30 to 54 years) and the Pittsburgh Imaging Project (PIP; N = 331; 50% women; age range = 30 to 51 years). Table 1 summarizes characteristics of the analytical sample derived from these cohorts. As detailed previously, both AHAB-2 (17) and PIP (33) had common recruitment and

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assessment protocols and both had harmonized assessments of behavioral, biological, and neural correlates of cardiometabolic risk in midlife adults (34). Participants in AHAB-2 and PIP provided informed consent prior to testing, and study approval was granted by The University of Pittsburgh Human Research Protection Office (AHAB-2 Protocol ID: 07040037; PIP Protocol ID: 07110287). Approval was also granted to merge AHAB-2 and PIP data (Protocol ID: 19030174). Recruitment for AHAB-2 and PIP entailed mass-mailings of study advertisements to residents of the greater Allegheny County area of Pennsylvania, USA. Addresses for mailings were derived from public domain lists (e.g., voter and vehicle registration registrations, etc.) between the years 2008 to 2011 for AHAB-2 and between 2008 to 2014 for PIP. Periods of data collection thus spanned from 2008 to 2014.

Common exclusions for AHAB-2 and PIP were any self-reported history of a diagnosed cardiovascular disease or clinical treatment for a cardiovascular disease; self-reports of a psychotic disorder, such as schizophrenia; chronic renal or hepatic diseases; history of seizure or cerebrovascular disorders; ongoing treatment for cancer or its treatment within the year prior to enrollment; chronic respiratory diseases; and, for women, pregnancy or recent pregnancy. Other common exclusions were (a) current usage of medications to control insulin, glucose, glucocorticoids, arrhythmias, blood pressure, lipids, weight, and mood at the time of screening; and (b) contraindications for brain imaging by magnetic resonance imaging (MRI; e.g., claustrophobia, metallic implants, etc.). Exclusion criteria unique to AHAB-2 included: resting blood pressure equaling or exceeding 160mmHg systolic and/or 100mmHg diastolic blood pressure (BP); consuming 5 or more alcoholic beverages 3+ times per week; taking supplements to increase fatty acid levels; routine shift-work; and less than part time employment outside of the home (< 25 hours per week). Exclusion criteria unique to PIP included: red-green color-blindness and having a resting blood pressure equaling or exceeding 140mmHg SBP and/or 90mmHg DBP. In AHAB-2, the Mini International Neuropsychiatric Interview (35) was used for mental health screening, whereas in PIP the Patient Health Questionnaire (36) was used for mental health screening.

Combining data from AHAB-2 and PIP yielded a total maximum possible sample of 821 individuals. Of these, there were 44 people who participated in both AHAB-2 and PIP. Only data obtained as part of AHAB-2 were used in the present report. Of the remaining 777 participants, several were missing data on one or more variables, including brain imaging outcome measures because of failure to acquire structural brain images, anatomical abnormalities detected on visual inspection, or poor-quality structural images unsuitable for processing. Missing data per variable ranged from 0% to just over 1%. To adjust for missing data, we implemented full information maximum likelihood estimation in Mplus (FIML) (37–39). FIML uses the information available for estimation, wherein cases with missing data on the outcome are retained in analyses if they have data on some predictors. Cases with missing values on independent variables were excluded prior to analysis. Along with the removal of data from the 44 PIP participants included in AHAB-2, this resulted in an analytic N = 259 for PIP and N = 440 for AHAB-2, yielding a final combined analytic sample size of N = 699.

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Overview of protocols to assess cardiometabolic risk and brain morphology

AHAB-2 and PIP involved multiple assessment visits, which have been detailed (17, 33, 34). Study visits entailed medical history and sociodemographic interviewing, as well as clinic protocols to measure seated resting blood pressure, anthropometrics (i.e., body mass index [BMI] and waist circumference), and fasting levels of glucose, high-density lipoproteins (HDL), total cholesterol, and triglycerides. MRI assessments in AHAB-2 and PIP were completed on the same MRI scanner at the same imaging center (3T Trio total imaging matrix scanner, Siemens, Erlangen, Germany), and used an identical T_1-weighted magnetization prepared rapid gradient echo (MPRAGE) sequence (7 min and 17 sec in duration) to acquire structural brain images (field-of-view = 256×208 mm, matrix size = 256 × 208, time-to-repetition = 2100 ms, inversion time = 1100 ms, time-to-echo = 3.31 ms, flip angle = 8°; 192 slices, 1 mm thickness, no gap).

Assessment of individual- and area-level indicators

Individual-level sociodemographic and socioeconomic indicators.—To examine the effects of community-level indicators apart from individual-level sociodemographic and socioeconomic factors, the following were assessed and included as *a priori* covariates in statistical models: chronological age (in years); self-reported sex at birth (coded as male = 1; female = 0); educational attainment (years of schooling completed); and annual, pretax income in US dollars (coded as <$10K; $10–14 999K; $15–24 999K; $25–34 999K; $35–49 999K; 50–64 999K; 65–79 999K; 80–94 999K; 95–109 999K; 110–124 999K; 25–139 999K; 140–154 999K; 155–169 999K; 170–185K; and >185K). Self-reported race/ethnicity (coded as white = 1; non-white = 0) were modeled as social constructs to account for the potential influence of inequalities in exposures to systemic racism and discrimination, as well as inequalities in access to opportunities among those who do not identify as white (40, 41).

Community-Level Indicators

Participants' street addresses were geocoded to derive community-level indicators for path analyses. Figure S1 (Supplemental Digital Content) shows a geocoded map that illustrates the distribution of address in the analytical sample. We utilized data from 4 public sources about residential area features at the Zone Improvement Plan (ZIP) code or census tract level. The first was the Child Opportunity Index 2.0 public dataset (COI; 42). It includes data obtained from third-party sources on several area-level features for almost all census tracts in the USA. Remaining sources included the USA Census Bureau, including the 2010 Decennial Census and ZIP Codes Business Patterns (ZBP), as well as the Federal Bureau of Investigation's (FBI's) Uniform Crime Reporting (UCR) database. These sources were utilized to create 8 different area-level indicators that capture unique features of communities (described below). Tables S1 and S2 (Supplemental Digital Content) provide cohort-specific descriptive statistics and bivariate correlations among the area-level indicators for the analytic sample, respectively.

A measure of concentrated area socioeconomic disadvantage was created as a composite of the following variables, obtained from the 2010 Decennial Census at the census tract level: 1) proportion of residents who lived under the federal poverty line; 2) percentage of

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residents receiving public assistance; 3) unemployment rate; 4) percentage of adult residents without a high school degree; and 5) percent of female-headed households (43, 44). These were standardized and averaged to create a composite score (α = .92).

Air pollution concentration, defined as the mean estimated concentration of particulate matter with aerodynamic diameter less than or equal to 2.5 μm (PM 2.5) within a census tract, was taken from the COI. This indicator incorporates data from the Centers for Disease Control (CDC) and Prevention and the Environmental Protection Agency (EPA), which use air monitoring data and simulation/prediction modeling to estimate PM 2.5 levels for all census tracts within the contiguous US for each day of the modeling year (42).

An indicator of risk posed to the public from toxic chemical releases, also available in the COI database, was created using the EPA's Toxic Release Inventory (TRI), specifically the Risk-Screening Environmental Indicators (RSEI) score (42). The EPA tracks pollution in the form of chemicals released into the air, water, and soil by industrial facilities, including measuring the release, movement, and chemical alteration of contaminants as they travel through air, soil or water, the size and location of the exposed population, and toxicity level of over 600 toxic chemicals. Using this information, RSEI scores are calculated by multiplying toxicity by the exposed population and the estimated dose of toxic chemicals (42).

We assessed area-level violent crime using precinct-level data on homicides obtained from the FBI's UCR database. The UCR collects monthly reports of crimes, separated by type and reported to police precincts. We calculated the residential-area homicide rate as the average number of monthly homicides per 10,000 people. Because police precincts often cover an area larger than the ZIP code in which they sit, and vice versa, not every ZIP code has its own police precinct. Accordingly, we aggregated homicide rates from all precincts within a 2-mile radius of each ZIP code.

COI created an indicator of lack of healthy food access defined as the proportion of households within the census tract without a vehicle that live further than a half-mile from the nearest supermarket. Data used to create this measure were taken from the Food Access Research Atlas created by the Department of Agriculture and Census Bureau's American Community Survey (ACS; 42).

A green space indicator was created using COI data, which drew information from the CDC on the percent of impenetrable surface areas (e.g., roads, rooftops, parking lots) within a census tract. We reverse coded this variable (i.e., multiplied by −1) so that the measure represents 'green space' within a census tract.

A residential-area indicator of the concentration of employment opportunities was drawn from an earlier version of the COI (45). This measure captured the average number of employers within 5 miles of the census tract. The COI standardized the measure.

Lastly, an aggregate indicator of area resources was created from data available from the ZBP resource of the Census Bureau. ZBP provides annual data on the number of business and service establishments in each ZIP code, broken out by type of establishment according

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to North American Industry Classification System codes (46). Using these data, we created a measure of resources across several sectors: medical/health resources (e.g., doctors' offices, hospitals, gyms); cultural attractions (e.g., museums, libraries, zoos, recreational facilities), educational services (e.g., tutoring services, schools), and other services (e.g., counseling services, social service agencies, drug and alcohol self-help centers). Different types of resources were highly intercorrelated and were not examined separately because of collinearity. As a result, the aggregate indicator represents the total number of resource services within a participant's ZIP code.

Cardiometabolic risk

Cardiometabolic risk factors were selected on the basis of their prior associations with community disadvantage and reductions in brain tissue volume among adults (e.g., 17, 20), as well as their known inter-relationships as encompassed by the broader construct of cardiometabolic risk (47). Seated resting blood pressures were derived from the average of 2 consecutive readings of SBP and DBP. BMI was computed as weight in kg per meters squared. Waist circumference was measured at end-expiration with a tape measure centered at the level of the umbilicus. After an overnight fast, samples of serum obtained by venipuncture were used to assay levels of lipids, and glucose. SBP, DBP, BMI, waist circumference, high-density lipoproteins (reverse-coded), triglycerides, and glucose were used to model overall cardiometabolic risk, as described previously (17). Triglyceride and fasting insulin levels were transformed by taking the natural log prior to further analysis. Scores on each variable were standardized and averaged to create a composite cardiometabolic risk variable (α = .82).

Brain tissue volumes

Volumetric brain imaging data were computed with FreeSurfer v6.0 (http://surfer.nmr.mgh.harvard.edu). MPRAGE images were submitted to the 'recon-all' pipeline, followed by manual inspection of processed data prior to computing total cortical tissue volume and total hippocampal volume as the sum of both hemispheres (48, 49). Separately, estimates of intracranial volume (ICV) were derived from the statistical parametric mapping toolbox, version 12 (SPM12) in MatLab (2018b) (Math-Works, Natick, MA, USA). Here, segmented gray matter, white matter, and cerebrospinal fluid tissue compartments were summed to compute ICV in mm squared (M = 6,040,000 ± 653,000 SD). Analyses were restricted to cortical and hippocampal tissue volumes produced by FreeSurfer owing to their relevance to midlife neurocognitive aging, dementia risk, and prior findings on these variables in association with community disadvantage (e.g., 17, 20). As noted below, analyses included ICV as a covariate in structural equation models to estimate *relative* tissue volume across individuals.

An indicator of image quality was used in ancillary sensitivity tests, as per methods described previously (50). In brief, MPRAGE images were submitted to the CAT12 toolbox (Computational Anatomy Toolbox 12) implemented in SPM12 to derive a single summary measure of image quality (51). For each image, the latter reflects an aggregate composite of noise, bias, and resolution on a percentage scale, wherein higher values (100% maximum) indicate better image quality. Sensitivity testing showed that removal of observations from

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10 individuals with image quality scores less than 85% and use of the image quality score as a covariate in the analyses reported below yielded similar findings and did not change the pattern of results reported below in direction or statistical significance. Accordingly, final statistical models included the full analytical sample of 699 participants and were not adjusted for image quality.

Data Analysis

Our first step was to explore the functional form of relations between community-level features and cardiometabolic risk. To do so, we examined plots of these relations and fit a lowess line to each association to determine the possible presence of nonlinear associations, as has been reported previously in studies of community disadvantage and neurodegenerative brain changes (e.g., 19, 20, 21). Any observed nonlinearities were modeled with the most appropriate nonlinear function based on the data. Next, we examined relations between cardiometabolic risk and brain tissue volumes. Finally, multi-level structural equation models (MSEMs) were estimated to test whether community features related to cortical and hippocampal tissue volumes statistically via cardiometabolic risk (i.e., as an indirect path). Due to the multi-level nature of the data, i.e. participants were clustered within 92 zip codes and 322 census tracts, we allowed the intercept to vary across areas by estimating a random intercept (71). Observations per zip code ranged from 1 to 59, with an average of approximately 8 and 33 zip codes with just 1 participant. There was less clustering within census tracts; observations per tract ranged from 1 to 15, with an average of approximately 2 and 114 tracts with just 1 participant.

The MSEM models were built sequentially. By this approach, we first estimated associations between community features and cardiometabolic risk simultaneously, controlling for individual-level indicators, so that coefficients reflected unique associations between community features and cardiometabolic risk. Next, we estimated a full path model to examine whether community features predicted cardiometabolic risk (a to b path), and cardiometabolic risk predicted brain volume (b to c path). Cortical volume and hippocampal volume were tested in separate models.

Models were estimated in Mplus (version 8) (39) using a Bayesian estimator (52). We evaluated the statistical significance of direct and indirect effects and correlations using posterior standard deviations and one-tailed p-values in line with a priori hypotheses based on the direction of associations (e.g., higher community disadvantage associating with increased cardiometabolic risk and reduced tissue volume) that have been reported previously (52). For completeness of reporting and to provide a fuller inferential context, two-tailed 95% confidence intervals (CI's) are presented in Tables 2–4, which summarize MSEM results. Due to the nonlinear relation between cardiometabolic risk and brain tissue volume (discussed in detail below), indirect effects were tested using the method described by Hayes and Preacher (53) for quantifying and testing indirect effects when the path between the mediator and the outcome (b to c) is nonlinear. A nonlinear association between cardiometabolic risk and brain tissue volume, in the context of this larger model testing mediation, means that the strength of the indirect effect between community features and brain tissue volume differs by level of cardiometabolic risk. Accordingly, we utilized the

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following equations to test indirect effects of community features on brain tissue volume through cardiometabolic risk at three levels of cardiometabolic risk (53):

1. theta1=a1*b1+a1*2*b2*predm1;
2. theta2=a1*b1+a1*2*b2*predm2;
3. theta3=a1*b1+a1*2*b2*predm3.

In these equations, the instantaneous indirect effect (theta) of community features on brain tissue volume through cardiometabolic risk is a function of the community feature multiplied by the linear cardiometabolic risk (a1*b1) plus the community feature multiplied by two-times the quadratic cardiometabolic risk term multiplied by the predicted value of the b to c association at 1 SD below the mean level cardiometabolic risk in the sample (predm1), at the mean level of cardiometabolic risk (predm2), and 1 SD above the mean level of cardiometabolic risk (predm3).

In addition to the *a priori* covariates noted above (age, sex, individual-level years of schooling, and annual income, and self-reported identification as white/non-white), paths predicting cortical and hippocampal volumes included ICV as an additional covariate. Finally, sensitivity tests were conducted to determine whether associations between variables of interest differed by study cohort (AHAB-2 vs. PIP) by interacting a cohort indicator with community features to predict cardiometabolic risk (a to b path) and with cardiometabolic risk to predict brain tissue volume b to c path.

Results

Are community features associated with cardiometabolic risk?

Analyses examining the functional form of relations between our main study variables showed several nonlinear associations between community features and cardiometabolic risk. More specifically, toxic chemical pollution, green space, and area resources all had nonlinear associations with cardiometabolic risk. Hence, threshold effects appeared for these relations, i.e., associations appeared to plateau or emerge at a certain point or threshold. Hence, toxic chemical pollution was unrelated to cardiometabolic risk until the toxic chemical index score reached 9, at which point increases in the toxic chemical index were related to higher cardiometabolic risk. A threshold effect for green space operated differently: increases in the amount of green space within an area were associated with lower rates of cardiometabolic risk until an area consisted of at least 40% green space. Once an area reached 40% green space, additional green space was unrelated to cardiometabolic risk. Lastly, the total number of area resources was unrelated to cardiometabolic risk until the area had at least 150 resources. After 150 resources, additional resources related to lower cardiometabolic risk. These nonlinear relations were best modeled by spline functions, which allow slopes to be estimated at different levels of the predictor variable (e.g., 54, 55). Accordingly, in path models, two continuous variables were included for toxic pollution, green space, and area resources: one representing the slope of relations between the community-level feature and cardiometabolic risk below the threshold, and one representing relations above the threshold. None of the other community-level features

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(PM 2.5, homicide rate, concentrated disadvantage, lack of healthy food access, and concentration of employment opportunities) exhibited nonlinear associations.

Table 2 shows standardized coefficients from models estimating relations between community features and cardiometabolic risk. Several of the community-level features related to cardiometabolic risk, while holding other community, individual-level socioeconomic, and demographic indicators constant. Toxic chemical releases, which as noted above had nonlinear relations with cardiometabolic risk, were positively related to cardiometabolic risk once the toxic chemical index surpassed 9 (which is about the mean level in the sample). At lower levels of risk from toxic chemicals, no associations were statistically evident between chemical releases and cardiometabolic risk. However, once the community-level toxic chemical risk index reached 9, increases in risk from toxic chemicals appeared associated with increased cardiometabolic risk (.11 *SD* increase in cardiometabolic risk per *SD* increase in the toxic chemical risk index). The community-level homicide rate was also positively related to cardiometabolic risk, such that participants living in areas with higher homicide rates had higher levels of cardiometabolic risk, on average (.10 *SD*). Similarly, community-level concentrated disadvantage had a positive association with cardiometabolic risk. Participants living in tracts with higher levels of concentrated disadvantage tended to have elevated cardiometabolic risk levels compared to those living in less-disadvantaged tracts (.10 *SD*). The percentage of green space within tracts was also related to participants' cardiometabolic risk. As green space increased, levels of cardiometabolic risk decreased (−.12 *SD*); however, this association was only evident until green space covered at least 40% of the tract, at which point further increases in green space were unrelated to cardiometabolic risk. Concentration of employment opportunities had a negative linear relation to cardiometabolic risk (−.16 *SD*), with those living in communities with more employment opportunities having lower cardiometabolic risk scores. Finally, the number of area-level resources within tracts was negatively related to cardiometabolic risk, but only once a certain number of resources was met. Hence, once tracts had at least 150 resources, additional resources were linked to less cardiometabolic risk (−.08 *SD*). Only 2 community-level features did not statistically relate to participants' cardiometabolic risk linearly or nonlinearly: PM 2.5 concentration and lack of access to healthy food.

Is cardiometabolic risk associated with brain tissue volumes?

Next tested was whether increased cardiometabolic risk was related to reduced cortical and hippocampal volumes. Initial scatterplots showed that the associations between cardiometabolic risk and cortical and hippocampal volume were best fit by a quadratic function, such that increases in cardiometabolic risk had stronger negative relations with cortical and hippocampal volumes at higher levels of cardiometabolic risk. These nonlinear associations were represented in the path models by including both linear and quadratic cardiometabolic risk variables as predictors. Table 3 presents standardized results of the linear and quadratic associations between cardiometabolic risk and tissue volumes (the b to c path) taken from the MSEM path models (cortical and hippocampal volume were separate models).

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As summarized in Table 3, the linear measure of cardiometabolic risk was unrelated to cortical and hippocampal tissue volumes. The quadratic cardiometabolic risk term, however, was significantly related to both cortical and hippocampal volumes, such that at higher levels of cardiometabolic risk, there was a stronger effect size of the negative association between cardiometabolic risk and tissue volumes.

Does cardiometabolic risk statistically mediate relations between community features and brain tissue volumes?

The final aim of this study was to explore whether community-level features had indirect effects on brain tissue volumes via cardiometabolic risk. Because cardiometabolic risk had nonlinear relationships with cortical and hippocampal volumes, indirect effects were tested at different values of cardiometabolic risk as per (53). Hence, we tested indirect effects of community-level features on brain tissue volumes at 1) one *SD* below the mean level of cardiometabolic risk, 2) the mean level of cardiometabolic risk, and 3) one *SD* above the mean level of cardiometabolic risk. Table 4 shows results of indirect effects tests at these values for the community features that were observed to statistically relate to cardiometabolic risk; namely, toxic chemical releases, homicide rates, concentrated disadvantage, green space, concentration of employment opportunities, and area resources.

With respect to cortical volume, results showed that at mean levels of cardiometabolic risk and at levels of cardiometabolic risk one *SD* above the mean, toxic chemical releases, homicide rate, and concentrated disadvantage all exhibited significant negative indirect effects on cortical volume. Stated differently, increased risk from toxic chemical releases, homicide rate, and concentrated disadvantage were all indirectly related to lower cortical volume, but these indirect effects were significant only in participants who had levels of cardiometabolic risk at mean levels or higher. Green space, concentration of employment opportunities, and area resources also had significant indirect effects, in the positive direction, on cortical volume when examining participants with cardiometabolic risks at mean levels or higher. Notably, some indirect effects for green space and area resources met directional (one-tailed) statistical thresholds, but had 95% two-tailed CI's that included 0.

Results with respect to the indirect effects of community features on hippocampal volume followed similar patterns of association as for cortical tissue volume, though associations were weaker in effect sizes and nearly all two-tailed 95% CI's included 0 (see Table 4).

In final sensitivity analyses, the indirect effects and corresponding models reported above were tested for effect modification by study cohort (AHAB-2 vs. PIP). Of all the community features tested, there was only one significant interaction, suggesting that associations between resources and cardiometabolic risk (i.e., an a path) appeared to be stronger in effect size in the AHAB-2 cohort (Table S3, Supplemental Digital Content).

Discussion

The present study examined whether features of residential communities that are not redundant with a conventional Census-based indicator of socioeconomic disadvantage (2) relate to presumptive brain correlates of neurocognitive risk via a mediational pathway

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encompassing risk factors for cardiometabolic diseases. Among an otherwise healthy and predominantly white sample of 699 midlife adults, we observed patterns of association that agree with and extend prior findings linking indicators of community socioeconomic disadvantage to elevated cardiometabolic risk and relative brain tissue reductions (17, 20, 21). Novel findings were that community features reflecting toxic pollution, homicide rates, green space, concentration of employment opportunities, and area resources, in addition to a conventional Census-based composite metric of socioeconomic disadvantage, exhibited indirect effects on cortical tissue volume via cardiometabolic risk. The latter findings were evident when all community features were modeled simultaneously and when accounting for individual-level socioeconomic and demographic factors. Several of these associations were nonlinear, and the same indirect effects were apparent for hippocampal volume, but the latter were weaker in magnitude and appeared less reliable, as nearly all two-tailed 95% CI's contained 0 (see Table 3). Notwithstanding the interpretive constraints of the present cross-sectional design and study limitations, these findings agree with the possibility that features of residential communities that are not redundant with those conventionally reflecting socioeconomic disadvantage may correspond to unique environmental correlates and possible nonlinear determinants of neurocognitive risk.

Educational, occupational, and financial dimensions of socioeconomic disadvantage measured at the level of communities (i.e., residential areas) have long been associated with elevated rates of cardiometabolic risk factors and their clinical sequelae (e.g., 2, 5, 6, 9, 56). In turn, cardiometabolic risk factors appear to confer risk for premature cognitive decline, adverse functional and structural changes in the brain, and several dementias (26, 28–31, 47, 57–63). Notably, community-level disadvantage itself has also been related to poorer cognitive function, dementia risk, and accelerated cognitive decline (15, 16, 21–25, 27), which may be attributable in part to cardiometabolic risk factors linked to premature brain aging and dementia risk. Recent evidence supporting the latter possibility indicates that community-level disadvantage in midlife is associated with reduced brain tissue volume, as well as reduced cortical surface area, cortical thickness, and white matter integrity (17, 18, 20, 21). Importantly, the associations of community disadvantage with brain morphology in prior studies and in the present study appear independent of individual-level socioeconomic factors and partially accounted for by cardiometabolic and related risk factors. To date, however, what has not yet been systematically tested is the extent to which social and physical features of residential communities that extend beyond conventional indicators of socioeconomic disadvantage may relate to presumptive brain correlates of neurocognitive risk via cardiometabolic pathways, as examined in the present study.

To elaborate, according to social epidemiological models, concentrated community socioeconomic disadvantage is thought to restrict access to material and social resources that impact physical health across the lifespan, as well as increase exposures to adverse systemic, social, and physical features of the residential environment that are detrimental to health (2, 12, 14). In these regards, models of community influences on health emphasize that apart from crude metrics of community socioeconomic disadvantage, there are nonredundant social and physical features of communities that may likewise operate as residential area determinants of health. Such features encompass aspects of social and built environments and access to resources that – in addition to concentrated disadvantage – may also contribute

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to the effects of communities on health and chronic diseases of aging. More specifically, communities characterized by higher levels of social discord may shape health behaviors and aspects of physiology that can confer health risk (2, 44). Results of the present study suggest that community homicide rates and opportunities for employment may relate to cardiometabolic risk and associated variation in brain tissue volume, possibly via behavioral and physiological pathways. With respect to cardiometabolic risk, the most often studied features of the built environment are those that may increase exposures to toxins, as well as those that restrict or enable community access to nutritious foods, amenities for physical activity, and green space, which may all have downstream effects on health behaviors and cardiometabolic outcomes (e.g., 2, 12, 14, 64–66). Features of the built environment examined in the present study included toxic chemical and air pollution, green space (land use patterns), and concentration of local resources. Here, we found that some of these aspects of the built environment (chemical pollution, green space, and local resources) exhibited nonlinear associations with cardiometabolic risk and threshold effects that varied by area feature. The latter may be explained by aspects of the greater Pittsburgh, PA area and the study sample composition, which was free of chronic illness and thus exhibited a restricted range of cardiometabolic risk compared to other samples. Contrary to expectations, two of the community-level indicators examined in this study did not predict participants' cardiometabolic risk for reasons that are not clear: PM 2.5 and relative lack of access to healthy foods. It is possible that limited variation in these area metrics as well as mismeasurement of the actual physical area of our participants' residential communities could explain these null findings. Moreover, it should be emphasized that the present null findings for these features do not provide strong evidence that these factors are unrelated to cardiometabolic heath and brain volume.

Inferentially, the general contextual (area-level) effects observed in this study were small, though not inconsistent with effect sizes for area effects observed in other studies (e.g., 67, 68). Intraclass correlations, for example, ranged from 1.5%−13% for census tracts, depending on the community feature. The small general contextual effects of residential areas are not surprising if health outcomes measured at the level of the person are assumed to be driven to a larger extent by individual-level factors, as opposed to more distal residential area features (69). Notably, however, we controlled for several relevant sociodemographic and individual-level factors that relate to cardiometabolic risk and brain morphology, including age, education, and income. Despite small effect sizes, the present observations appear compatible with interpreting contextual influences on study outcomes.

Interpretations of the present findings are constrained by several limitations. First, the study sample was predominantly white and relatively advantaged, which limit generalizability to other samples and communities outside of a mostly urban region of western Pennsylvania. The latter limitations also restrict inferences regarding the intersection of race, ethnicity, and community context with respect to brain health. In these regards, there is a growing need to address and understand how unjust and racist historical housing and related policies that have created systemic influences on residential tenure, environmental exposures, psychosocial factors, and access to resources and opportunities may interactively influence brain aging and dementia risk (e.g., 70, 71, 72). Second, it is possible that unmeasured psychosocial (e.g., subjective perceptions of residential communities) and

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biobehavioral factors (e.g., measured engagement in physical activity, sleep, etc.) may also partly account for the influence of community features on cardiometabolic risk and brain morphology. Third, the present cross-sectional findings prevent causal inference and determining the temporal ordering of patterns of association. In the latter regards, however, it is noteworthy that Hunt and colleagues provided evidence for longitudinal associations between community disadvantage and neurodegenerative changes that statistically explained correlated changes in cognitive decline (21). Fourth, the present study was not statistically powered to test for the synergistic effects of community features or person-environment interactions, and it is plausible that some community features may operate together in nonadditive ways. Fifth, the present findings do not permit inferences regarding the possible neuropsychological (e.g., executive function, memory, etc.) changes associated with brain tissue volume reductions, as well as the possible cellular mechanisms or neuropathological changes (e.g., 26) that may account for reduced tissue volumes across participants. Sixth, we note that interpretive weight should be stronger for findings in Tables 2–4 where hypothesis-driven directional (1-tailed) statistical testing thresholds were accompanied by 2-tailed 95% CI's that did not include 0, as as more evident for cortical tissue volume than for hippocampal volume. Lastly, it was not possible to test for the influence of housing tenure (length of residence) or early life area-level exposures that appear to also relate to relative changes in brain tissue morphology earlier than midlife (e.g., 73, 74).

In aggregate, the present findings extend accumulating evidence on community features and brain changes linked to risk for chronic health conditions, which to date have largely focused on individual-level dimensions of socioeconomic disadvantage or nonspecific markers of area disadvantage (for reviews, see 75, 76). The present findings suggest that multiple community (area-level) features apart from individual-level characteristics associate with cardiometabolic risk factors and associated changes in brain morphology that may presage emergent cognitive outcomes in later life. In these regards, the present findings underscore the importance of extending the measurement of community socioeconomic contexts to include features beyond global metrics of concentrated disadvantage, including features of the social and built environments that may shape brain aging.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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List of Abbreviations:

ACS	American Community Survey
AHAB-2	Adult Health and Behavior, Phase 2

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BMI	body mass index
BP	blood pressure
CAT	Computational Anatomy Toolbox
CDC	Centers for Disease Control and Prevention
COI	Child Opportunity Index
EPA	Environmental Protection Agency
FBI	Federal Bureau of Investigation
FIML	full information maximum likelihood estimation
HDL	high-density lipoproteins
ICV	intracranial volume
ID	identifier
mmHg	millimeters of mercury
MPRAGE	magnetization prepared rapid gradient echo
MSEM	multi-level structural equation model
PIP	Pittsburgh Imaging Project
PM	particulate matter
RSEI	Risk-Screening Environmental Indicators
SPM	statistical parametric mapping
TRI	Toxic Release Inventory
UCR	Uniform Crime Reporting
USA	United States of America
ZBP	ZIP Codes Business Patterns
ZIP	Zone Improvement Plan

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Table 1.

Descriptive Statistics for the Analytical Sample (N = 699)

Variable	Mean / %	(*SD*)	Min	Max
Cardiometabolic risk	−0.01	(0.64)	−1.75	2.71
Cortical volume	0.02	(0.99)	−2.44	2.85
Hippocampal volume	0.01	(1.00)	−3.28	3.76
Community Features:				
Particulate matter (PM2.5)	13.23	(0.31)	11.17	13.54
Toxic chemical releases risk	9.30	(0.53)	7.54	11.62
Homicide rate	0.04	(0.05)	0	.25
Concentrated disadvantage	−0.02	(0.84)	−0.97	3.86
Lack of healthy food access	0.26	(0.97)	−1.10	2.51
Green space	59.32	(20.21)	15.10	99.50
Concentration of employment opportunities	0.77	(1.08)	−0.76	2.59
Area resources	93.75	(55.97)	0	237
Individual-Level Demographic Factors:				
Years of schooling	16.91	(2.97)	9	24
Income (scaled in $10,000 increments)	71.55	(44.41)	5	185
Age (years)	41.57	(7.09)	30	54
Sex at birth (male)	48.6%		0	1

Note. N = 699. Cardiometabolic risk is a composite of standardized variables. Brain tissue volumes are in standardized units. Particulate matter concentration is measured as $\mu g/m^3$. Toxic chemical releases are computed as RSEI scores, wherein a toxicity weight is multiplied by the exposed population multiplied by the estimated dose based on emission data of over 600 toxic chemicals. Homicide rates are measured as homicides per 10,000 people (% on 0–1 scale). Concentrated disadvantage, lack of healthy food access, and concentration of employment opportunities are comprised of standardized variables. Green space is measured in percent (1–100), and area resources are measured as a total count variable. Participants who self-identified as non-white (22.5%) included 64 (AHAB-2) and 66 (PIP) individuals identifying as Black (African American); 6 (AHAB-2) and 14 (PIP) individuals identifying as Asian or Pacific Islander; and 3 (AHAB-2) and 4 (PIP) individuals identifying as multiracial or other. There were 8 (AHAB-2) and 5 (PIP) individuals identifying as Hispanic or Latino/a/x. In multilevel structural equation models, there were 87 unique zip codes and 313 tracts represented in the sample.

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Table 2.

Community Features Predicting Cardiometabolic Risk

	Cardiometabolic Risk		
	B	**Posterior *SD***	**95% Cis Lower | Upper**
Community Features:			
Particulate Matter (PM 2.5)	0.06	(0.05)	−0.04 | 0.15
Toxic chemical releases risk < 9	−0.04	(0.04)	−0.12 | 0.04
Toxic chemical releases risk ≥ 9	0.11**	(0.04)	0.02 | 0.19
Homicide rate	0.10*	(0.04)	0.01 | 0.18
Concentrated disadvantage	0.10**	(0.04)	0.01 | 0.18
Lack of healthy food access	0.04	(0.04)	−0.04 | 0.11
Green space < 40%	−0.12**	(0.04)	−0.20 | −0.04
Green space ≥ 40%	0.02	(0.06)	−0.09 | 0.14
Concentration of employment opportunities	−0.16**	(0.06)	−0.27 | −0.04
Area resources < 150	0.06	(0.04)	−0.02 | 0.14
Area resources ≥ 150	−0.08*	(0.04)	−0.16 | 0.00
Covariates:			
Years of schooling	−0.11**	(0.04)	−0.18 | −0.04
Income	−0.02	(0.04)	−0.10 | 0.06
Age	0.12***	(0.04)	0.06 | 0.19
Sex at birth (male)	0.36***	(0.03)	0.30 | 0.42
Self-identification as white vs. non-white	−0.08*	(0.04)	−0.15 | 0.00

N = 699.

*** $p < .001$

** $p < .01$

* $p < .05$.

Factors shown above with two predictors represent factors measures with non-linear spline functions at the knot point specified above.

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Table 3.

Cardiometabolic Risk Predicting Cortical and Hippocampal Volumes

	Cortical Volume			Hippocampal Volume		
	β	**Posterior *SD***	**95% Cis Lower | Upper**	**β**	**Posterior *SD***	**95% Cis Lower | Upper**
Cardiometabolic risk (linear)	−0.03	(0.02)	−0.07 | 0.01	0.02	(0.03)	−0.05 | 0.08
Cardiometabolic risk^2 (quadratic)	−0.07***	(0.02)	−0.11 | −0.04	−0.08**	(0.03)	−0.14 | −0.02
Intracranial volume	0.69***	(0.02)	0.65 | 0.73	0.47***	(0.03)	0.40 | 0.53
Years of schooling	−0.002	(0.02)	−0.04 | 0.04	0.00	(0.03)	−0.06 | 0.06
Income	0.04*	(0.02)	0.00 | 0.08	0.03	(0.03)	−0.03 | 0.10
Age	−0.22***	(0.02)	−0.26 | −0.19	−0.08**	(0.03)	−0.14 | −0.02
Sex at birth (male)	0.13***	(0.03)	0.08 | 0.18	0.13**	(0.04)	0.06 | 0.21
Self-identification as white vs. non-white	0.12***	(0.02)	0.08 | 0.16	0.15***	(0.03)	0.09 | 0.21

N = 699.

*** $p < .001$

** $p < .01$

* $p < .05$

Covariates included age, sex at birth, individual-level years of schooling, annual income, and self-identification as white/non-white as predictors of cardiometabolic risk and brain tissue volume. The covariate of intracranial volume was included as an additional predictor of tissue volume.

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Table 4.

Indirect Effects of Community Features on Brain Tissue Volumes at Levels of Cardiometabolic Risk

	Cortical Volume			Hippocampal Volume		
Indirect Effect	**Coeff.**	**Posterior *SD***	**95% Cis Lower | Upper**	**Coeff.**	**Posterior *SD***	**95% CIs Lower | Upper**
Toxic chemical releases risk ≥ 9:						
1 SD below mean cardiometabolic risk	0.003	(0.020)	−0.026 | 0.054	−0.012	(0.025)	−0.018 | 0.081
Mean cardiometabolic risk	−0.027*	(0.018)	−0.071 | −0.003	−0.017†	(0.018)	−0.065 | 0.006
1 SD above mean cardiometabolic risk	−0.030*	(0.018)	−0.075 | −0.005	−0.020†	(0.021)	−0.073 | 0.009
Homicide rate:						
1 SD below mean cardiometabolic risk	0.022	(0.154)	−0.011 | 0.481	0.091	(0.205)	−0.072 | 0.701
Mean cardiometabolic risk	−0.192*	(0.136)	−0.539 | −0.012	−0.117†	(0.138)	−0.486 | 0.038
1 SD above mean cardiometabolic risk	−0.426*	(0.352)	−1.359 | −0.022	−0.351*	(0.378)	−1.415 | −0.002
Concentrated disadvantage:						
1 SD below mean cardiometabolic risk	0.001	(0.009)	−0.015 | 0.022	0.005	(0.011)	−0.010 | 0.034
Mean cardiometabolic risk	−0.012*	(0.008)	−0.033 | −0.001	−0.007†	(0.008)	−0.030 | 0.003
1 SD above mean cardiometabolic risk	−0.026*	(0.018)	−0.070 | −0.002	−0.022*	(0.019)	−0.073 | 0.000
Green space < 40%:						
1 SD below mean cardiometabolic risk	0.000	(0.002)	−0.004 | 0.003	−0.001	(0.002)	−0.005 | 0.002
Mean cardiometabolic risk	0.003**	(0.002)	0.000 | 0.008	0.002†	(0.002)	0.000 | 0.007
1 SD above mean cardiometabolic risk	0.006**	(0.004)	0.001 | 0.016	0.005*	(0.004)	0.000 | 0.016
Employment opportunities:						
1 SD below mean cardiometabolic risk	−0.001	(0.011)	−0.027 | 0.018	−0.007	(0.013)	−0.041 | 0.013
Mean cardiometabolic risk	0.016**	(0.011)	0.002 | 0.044	0.010†	(0.011)	−0.003 | 0.040
1 SD above mean cardiometabolic risk	0.034**	(0.023)	0.006 | 0.093	0.028*	(0.025)	0.001 | 0.097
Area resources ≥ 150:						
1 SD below mean cardiometabolic risk	0.000	(0.000)	−0.001 | 0.001	0.000	(0.000)	−0.001 | 0.000
Mean cardiometabolic risk	0.001*	(0.000)	0.000 | 0.001	0.000	(0.000)	0.000 | 0.001
1 SD above mean cardiometabolic risk	0.001*	(0.001)	0.000 | 0.003	0.001*	(0.001)	0.000 | 0.003

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N = 699.

*** $p < .001$

** $p < .01$

* $p < .05$

† $p < .10$

Covariates included age, sex at birth, individual-level years of schooling, annual income, and self-identification as white/non-white as predictors of cardiometabolic risk and brain tissue volume. The covariate of intracranial volume was included as an additional predictor of tissue volume.